

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. We construct projectively compatible positive phase labels from peripheral positive bands of the microscopic platform operator, including an explicit complementary label. Normalized component pressures, phase-local LDPs, exact mixture rates, shell coefficients, Gaussian tangents, and dynamic semigroups are assembled on one joint labelled law. Phase costs are counted once and label contraction yields the nonconvex minimum rate.

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1. ROUND-TWELVE POSITIVE CLOSURE: CANONICAL PHASE SHEAVES AND LOCALIZED MIXTURE LDPs

The label is not an arbitrary partition of unity. It is constructed from the finite family of positive peripheral bands of the platform transfer/cluster operator and is compatible under every finite projection. An explicit “overlap” label is retained. Normalized conditional pressures and unnormalized restricted partition functions are never conflated.

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1.1. Canonical positive phase bands. Let $\mathcal{T}_{n,\theta}$ be the positive finite-volume transfer/cluster operator of the platform on the declared regular coexistence chart. Assume its peripheral spectrum consists of finitely many isolated positive bands separated by a uniform annular gap; A2–A4 and B1–B4 provide these band operators on their respective charts. Positivity gives band projections $E_{n,j} \geq 0$. Put

$$\chi_{n,j}(x) = \frac{E_{n,j}1(x)}{\sum_k E_{n,k}1(x) + R_n1(x)}, \quad \chi_{n,\partial} = \frac{R_n1}{\sum_k E_{n,k}1 + R_n1},$$

where R_n is the positive complementary band. Thus the overlap/remainder label ∂ is an actual component.

Theorem 1.1 (Exact canonical disintegration). *The variables $J_n \in \{1, \dots, J, \partial\}$ with conditional probabilities $\chi_{n,j}$ define an exact positive disintegration of the original microscopic law. The same J_n is used for all finite projections, hence the labelled laws are projectively compatible. On the regular chart the ∂ weight is superexponentially small; it is nevertheless retained in every exact identity.*

Proof. Positive band projections and the complementary positive operator give a partition of unity pointwise. Sample the label conditionally on the original microstate; summing labels reconstructs the law exactly. Since the label is defined before applying any observable projection, compatibility is automatic. The peripheral gap and the platform compact-containment estimate make the complementary band superexponentially small on the chart. $\square$

1.2. Normalized component pressures. Let $w_{n,j} = \mathbb{E}\chi_{n,j}$ and $\mathbb{P}_{n,j} = w_{n,j}^{-1}\chi_{n,j}\mathbb{P}_n$. Define

$$\tilde{Q}_{n,j}(\theta) = \frac{1}{n} \log \mathbb{E}_{\mathbb{P}_{n,j}} e^{n\theta X_n}, \quad \alpha_j = -\lim \frac{1}{n} \log w_{n,j}.$$

The unnormalized restricted pressure is $Q_{n,j} = n^{-1} \log w_{n,j} + \tilde{Q}_{n,j}$.

Theorem 1.2 (Localized zero-free and component LDP theorem). *Each positive band has a phase-local zero-free complex chart, a full good conditional LDP with rate I_j , boundary-face recovery, and the uniform local coefficients/Gaussian tangents supplied by its platform theorem. The localized operator is the positive band operator itself, not multiplication by an arbitrary sharp basin indicator.*

Proof. The Riesz contour defining a positive peripheral band is part of the platform spectral/cluster operator. Restriction to its invariant range preserves the simple leading eigenvalue and its zero-free neighborhood. The A3 or B2 Laplace principles apply with the positive band as an additional finite state; their entropy-control recoveries can be conditioned on that state. Normal source convergence gives local coefficients and Gaussian derivatives.

Boundary faces are obtained by the same band-preserving normal-source limits. $\square$

Theorem 1.3 (Joint labelled LDP and exact phase minimum). *The pair (J_n, X_n) satisfies the good LDP*

$$\mathcal{J}(j, x) = \alpha_j + I_j(x), \quad \mathcal{J}(\partial, x) = I_\partial(x).$$

After contraction of the label,

$$I(x) = \min_j \{ \min(\alpha_j + I_j(x)), I_\partial(x) \}.$$

The physical pressure is

$$Q(\theta) = \max_j \{ \max[-\alpha_j + \tilde{Q}_j(\theta)], Q_\partial(\theta) \};$$

no phase cost is counted twice and no convexification is attributed to label contraction.

Proof. The upper bound is a finite union of the component upper bounds. For an open set choose a component attaining the infimum and use its conditional lower bound together with the asymptotic weight. Exponential tightness is uniform across the finite labels. Contraction gives the minimum rate. The exact partition-function sum gives the maximum pressure with the displayed normalization. $\square$

1.3. Conditioning, tangents, and dynamics.

Theorem 1.4 (Phase-compatible shell and commutation theorem). *For every regular shell, the exact joint labelled coefficient is the finite sum of the A2/B1 phase coefficients, including the ∂ component. Shell conditioning, typed observable contraction, Gaussian tangent formation, and the component dynamic semigroups commute on the labelled law. At coexistence every subsequential physical tangent is the mixture selected by the actual subexponential component weights; if these weights converge, the mixture is unique.*

Proof. Apply the phase-local coefficient before summing the exact positive disintegration. Boundary contributions are already part of the ∂ label, so no unlabelled overlap is lost. Conditioning and observable projection are ordinary conditional operations on one joint law. Phase-wise Gaussian and semigroup convergence are uniform on the finite band family, permitting finite summation. Tightness of the normalized weights gives subsequential mixtures, and convergence of the weights gives uniqueness. $\square$

Proposition 1.5 (Projective phase sheaf). *On overlaps of two regular source charts, positive band labels are identified by continuation of their Riesz contours. These identifications satisfy the cocycle law and form a finite phase sheaf over the regular pressure atlas. All finite observable projections use its pullback; labels are never redefined projection by projection.*

Proof. A peripheral band cannot cross another band without leaving the regular atlas. Analytic continuation of its contour is therefore unique on overlaps. Uniqueness implies the cocycle law. Pullback along an observable projection leaves the microscopic label unchanged. $\square$

Theorem 1.6 (Round-twelve D1 closure). *The Sinai and hard-sphere regular coexistence charts possess canonical positive phase labels, normalized component pressures, full component LDPs, exact minimum-rate contraction, and the phase-compatible shell/tangent/semigroup commutation theorem.*

Proof. Combine Theorems 1.1, 1.2, 1.3, and 1.4, with Proposition 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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